Report 1: Project Description Report

Problem Statement & Purpose

Traditional hospital workflows heavily rely on manual paperwork for patient registration, scheduling, and record-keeping. This often leads to misfiled medical records, long queues, scheduling conflicts, and delayed billing. The purpose of the **Hospital Management System** is to digitize these operations, ensuring secure data management, real-time appointment booking, quick access to patient medical histories, and transparent billing.

Project Scope

- **Functions Covered:**
  - Patient registration and profile management.
  - Staff (Doctor and Receptionist) management.
  - Appointment scheduling, confirmation, and cancellation.
  - Creation and management of medical records/prescriptions.
  - Invoice generation and payment processing.

- **Functions Excluded**
    - External insurance third-party API payment gateways (simulated locally).
    - Advanced pharmacy inventory tracking and supply chain management (handled via basic prescription logs instead).
    - In-patient room/bed allocation tracking (focused primarily on outpatient consultations and records).

Main Classes & Responsibilities

1. **Person (Abstract/Base Class):** Acts as a superclass holding shared attributes for all human actors in the system.
2. **Patient:** Represents individuals seeking medical care; manages personal health data and appointment history.
3. **Doctor:** Represents medical professionals; manages availability, diagnoses, and prescriptions.
4. **Receptionist:** Front-desk staff responsible for registering patients and booking appointments.
5. **Appointment:** Manages the scheduling link between patients, doctors, and receptionists.
6. **Medical Record:** Stores clinical notes, diagnoses, and prescribed treatments for patients.
7. **Billing:** Manages consultation fees, hospital charges, and payment statuses.